A Governance-First Agentic Method for Quantum Portfolio Optimization: From Discrete Portfolios to Variational Manifolds

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Abstract

This paper develops a governance-first methodology for quantum portfolio optimization using an agentic Google Colab workflow. The objective is not to claim near-term quantum advantage, but to present a rigorous and auditable architecture for translating a constrained portfolio-selection problem into a variational quantum optimization experiment. Candidate portfolios are encoded as computational-basis states, and a diagonal cost Hamiltonian assigns each state an energy determined by expected return, covariance risk, and explicit constraint penalties. A parameterized circuit then induces a low-dimensional variational family of probability distributions over the full combinatorial portfolio universe. The Hamiltonian expectation value becomes a differentiable surrogate loss that can be optimized with parameter-shift gradients and classical updates. The paper emphasizes the distinction between the financial model, the binary encoding, the Hamiltonian representation, the reachable variational manifold, the measurement distribution, and the decoded portfolio decision. It argues that an agentic architecture is particularly valuable in financial settings because it decomposes a complex quantum workflow into separately validated responsibilities: data generation, objective construction, Hamiltonian assembly, circuit design, optimization, sampling, decoding, benchmarking, and explanation. The discussion is aimed at readers with graduate-level finance and computer-science training and connects the portfolio example to broader quantum-finance applications, including amplitude estimation, risk aggregation, derivative pricing, credit analysis, and hybrid quantum-classical learning systems.

Colab notebook: Governance-First Agentic Quantum Portfolio Optimization Notebook

1 Introduction and Motivation

Portfolio optimization is a canonical example of a financial problem in which a compact economic objective can conceal substantial computational and governance complexity. At the portfolio-manager level, the task is familiar: select assets and assign exposures so as to trade off expected return against risk. At the institutional level, however, the optimization must also respect investment mandates, trading frictions, model-risk controls, explainability requirements, and independent validation. In the classical mean–variance framework, the continuous baseline problem can be written as

$$\min_{w} \quad \lambda w^{\top} \Sigma w - (1-\lambda)\mu^{\top} w,$$

subject to constraints such as

$$\sum_i w_i = 1, \qquad w_i \geq 0.$$

Here, w is the vector of portfolio weights, μ is the vector of expected returns, Σ is the covariance matrix, and $\lambda \in [0,1]$ controls the risk–return trade-off. In practice, μ is usually the least stable input, Σ requires careful estimation or shrinkage, and the objective should be interpreted as a model of preferences rather than as a revealed truth about future returns.

In its unconstrained or convex form, this problem is well understood and efficiently solvable with classical methods. The computational difficulty emerges when realistic institutional restrictions are added. These may include cardinality constraints, minimum and maximum position sizes, discrete trading lots, turnover limits, sector caps, concentration constraints, transaction costs, tax considerations, benchmark tracking, liquidity limits, exclusion lists, and conditional rules. The decision variables may become binary or mixed-integer, and the feasible set may become nonconvex. A portfolio problem with n binary inclusion decisions contains

$$2^n$$

possible bitstrings. With 20 assets, there are

$$2^{20} = 1{,}048{,}576$$

possible inclusion patterns. With 50 assets, the number of alternatives is approximately

$$1.13 \times 10^{15}.$$

The challenge is therefore not merely to evaluate one candidate portfolio, but to search a combinatorial decision space whose size grows exponentially while preserving economically meaningful constraints and auditability.

The notebook studied in this paper addresses this challenge through a pedagogical quantum representation. Each asset is associated with one binary variable

$$x_i \in \{0, 1\},$$

and a portfolio is represented by

$$x = (x_1, x_2, \ldots, x_n).$$

If a fixed cardinality K is imposed and selected assets receive equal weights, then

$$w_i = \frac{x_i}{K}.$$

The financial objective can be written as

$$C(x) = \lambda w^\top \Sigma w - (1 - \lambda)\mu^\top w + A \left(\sum_i x_i - K \right)^2,$$

where the penalty parameter A discourages violations of the desired number of selected assets. This penalty formulation is a standard unconstrained-binary-optimization device: feasibility is not removed from the state space, but infeasible bitstrings are made energetically unattractive. The size of A is therefore a governance parameter, because too small a value can permit economically invalid portfolios to dominate the sampled distribution, while too large a value can distort numerical conditioning and optimization dynamics.

The first major transformation is to encode each candidate portfolio as a computational-basis state

$$|x\rangle = |x_1 x_2 \cdots x_n\rangle.$$

The complete n-qubit Hilbert space contains one basis vector for each of the 2^n bitstrings. A cost Hamiltonian H_C is then constructed so that

$$H_C |x\rangle = C(x)|x\rangle.$$

The objective value of each portfolio becomes the eigenvalue associated with its basis state.

In the simplest educational implementation, the Hamiltonian is diagonal:

$$H_C = \sum_x C(x)|x\rangle\langle x|.$$

The spectrum of the operator is therefore a structured representation of the encoded opportunity set: each basis vector is a portfolio hypothesis and each eigenvalue is its risk–return–penalty score.

It is important to state precisely what this Hamiltonian does and does not accomplish. It does not reduce the number of portfolios. The basis still contains 2^n alternatives. It does not make the original discrete function differentiable with respect to the binary variables. Rather, it gives every candidate a common energy interpretation and makes it possible to evaluate a quantum state through the expectation value

$$\langle H_C \rangle.$$

The lowest-energy basis state represents the best portfolio under the encoded assumptions, not necessarily the best portfolio in an economic or fiduciary sense. That distinction matters: errors in return estimation, covariance estimation, constraints, or penalty calibration are faithfully inherited by the Hamiltonian.

The second major transformation is variational. A parameterized circuit prepares a state

$$|\psi(\theta)\rangle = U(\theta)|0\cdots 0\rangle,$$

where

$$\theta = (\theta_1, \ldots, \theta_p)$$

is a vector of continuous gate parameters. The state can be expanded as

$$|\psi(\theta)\rangle = \sum_x \alpha_x(\theta)|x\rangle.$$

The measurement probability of candidate x is

$$P_\theta(x) = |\alpha_x(\theta)|^2.$$

The parameterized circuit therefore defines a family of probability distributions over the full combinatorial solution space.

The set of states reachable by the chosen circuit is the variational manifold

$$\mathcal{M} = \{|\psi(\theta)\rangle : \theta \in \mathbb{R}^p\}.$$

This manifold is generally much lower-dimensional than the complete Hilbert space. A system with n qubits has 2^n computational-basis amplitudes, whereas the circuit may contain only p trainable angles. The circuit therefore does not independently control every amplitude. It imposes an inductive bias through gate choice, depth, parameter sharing, and entangling topology. In machine-learning language, the ansatz defines the hypothesis class; in optimization language, it defines the subset of distributions the algorithm can represent.

This lower-dimensional representation is central to the intuition of variational quantum optimization. The original problem asks for

$$\min_{x\in\{0,1\}^n} C(x).$$

The variational problem asks for

$$\min_{\theta\in\mathbb{R}^p} E(\theta),$$

where

$$E(\theta) = \langle\psi(\theta)|H_C|\psi(\theta)\rangle.$$

For a diagonal Hamiltonian,

$$E(\theta) = \sum_x P_\theta(x)C(x).$$

The expected energy is a probability-weighted average of the costs of all portfolios represented by the state.

This equation reveals the central conceptual move. The variational model does not move directly from one admissible portfolio to another. It changes a vector of circuit parameters, and those changes can reallocate probability mass across many portfolios simultaneously. The search is therefore conducted over parameterized distributions rather than isolated bitstrings. The resulting parameter landscape is differentiable under the usual smooth gate constructions even though the terminal portfolio produced by measurement remains discrete.

The method may therefore be understood as a quantum probabilistic relaxation of a discrete problem. It differs from the classical relaxation

$$x_i \in \{0,1\} \longrightarrow x_i \in [0,1].$$

In a classical relaxation, an intermediate point may represent a fractional portfolio. In the quantum variational approach, the intermediate state represents a probability distribution over discrete portfolios. Measurement still produces a discrete bitstring.

The circuit architecture determines the shape of the variational manifold. Independent one-qubit rotations generate factorized distributions. If

$$|\psi\rangle = |\psi_1\rangle \otimes \cdots \otimes |\psi_n\rangle,$$

then

$$P(x_1, \ldots, x_n) = \prod_i P_i(x_i).$$

This representation cannot capture complex dependencies among asset decisions. Entangling gates introduce nonfactorizable structure, allowing the model to express patterns analogous to joint inclusion, mutual exclusion, sector clustering, concentration effects, or nonlinear interactions among positions. Entanglement should not be romanticized as a guarantee of performance; its importance here is representational, because it expands the family of distributions beyond independent Bernoulli selections.

The Hamiltonian and the manifold therefore play different roles. The Hamiltonian gives the candidate set an objective structure. The variational circuit gives the algorithm a continuous coordinate system for navigating a structured family of probability distributions. The optimizer then uses measurements of the Hamiltonian to determine which parameter updates lower expected cost.

The notebook uses the parameter-shift rule. For suitable rotation gates,

$$\frac{\partial E}{\partial \theta_j} = \frac{1}{2}\left[E\left(\theta + \frac{\pi}{2}e_j\right) - E\left(\theta - \frac{\pi}{2}e_j\right)\right].$$

Gradient descent updates parameters through

$$\theta^{(t+1)} = \theta^{(t)} - \eta \nabla E(\theta^{(t)}),$$

where η is a learning rate.

The process is hybrid:

$$\begin{aligned}\text{classical parameters} &\to \text{quantum state} \to \text{Hamiltonian measurement}\\ &\to \text{classical parameter update.}\end{aligned}$$

This architecture is structurally similar to a quantum neural network. Rotation angles play the role of trainable weights, gate layers define the model class, expectation values provide outputs, and the Hamiltonian supplies the loss. The analogy is useful but limited: unlike most supervised learning systems, the training signal here is generated by an explicitly encoded optimization objective rather than by labeled examples.

The motivation for a governance-first implementation is that a numerically attractive result is not sufficient in financial decision systems. Every stage must be explainable, reproducible, and subject to controls. The data may be biased, the covariance matrix may be ill-conditioned, the penalty may be insufficient, the Hamiltonian may encode the wrong economics, the ansatz may be too shallow to reach high-quality states, the optimizer may stop at a poor stationary point, and the sampled bitstrings may violate mandate constraints. A professional workflow must expose these failure modes rather than bury them inside a single optimization routine.

The notebook therefore decomposes the process into agents with separation of concerns. A Governance Agent defines configuration and controls. A Data Agent creates and validates the return history. A Problem Encoding Agent calculates expected returns, covariance, and the financial objective. A Hamiltonian Agent constructs the energy spectrum. A Variational Manifold Agent defines the parameterized circuit. An Energy Measurement Agent evaluates expected cost and feasible probability. A Gradient Descent Agent navigates the manifold. A Sampling Agent produces candidate bitstrings. A Decoder Agent converts them into portfolios. A Reporter Agent compares the result against an exact benchmark and assembles an audit record. An LLM Explanation Agent summarizes the process without changing the underlying computations.

This agentic decomposition transforms the optimization from a monolithic procedure into a chain of governed claims and testable interfaces. The Data Agent claims that the observations are valid. The Problem Encoding Agent claims that the objective reflects the intended financial trade-off. The Hamiltonian Agent claims that each bitstring receives the correct cost. The Variational Manifold Agent claims that the circuit remains normalized. The Energy Agent claims that the expectation value is calculated correctly. The Gradient Agent claims that updates follow the declared rule. The Decoder claims that the final bitstring corresponds to a feasible portfolio. The Reporter claims that the result has been compared against the benchmark.

The educational value is substantial. The learner can observe the entire conceptual transformation:

$$
\begin{aligned}
\text{financial data} \to \text{discrete objective} \to \text{Hamiltonian} &\to \text{variational state} \\
&\to \text{continuous energy landscape} \to \text{gradient descent} \\
&\to \text{measurement} \to \text{portfolio.}
\end{aligned}
$$

This sequence reveals that quantum optimization is not a mysterious shortcut. It is a carefully engineered representation and search process.

The notebook uses an exact NumPy statevector simulation. This choice keeps amplitudes, probabilities, energies, and sampled bitstrings visible, and it avoids hiding the logic behind a high-level quantum software stack. It does not establish practical quantum advantage, hardware scalability, or production readiness. It does, however, provide a rigorous architecture for understanding how quantum ideas can be translated into financial algorithms and how such algorithms can be governed.

The broader motivation is that future quantum-financial systems will need more than speed. They will need model lineage, reproducibility, validation, explainability, and accountability. A quantum algorithm that produces a portfolio without preserving the path from data to decision would be difficult to use responsibly. The agentic governance-first methodology addresses this requirement from the beginning.

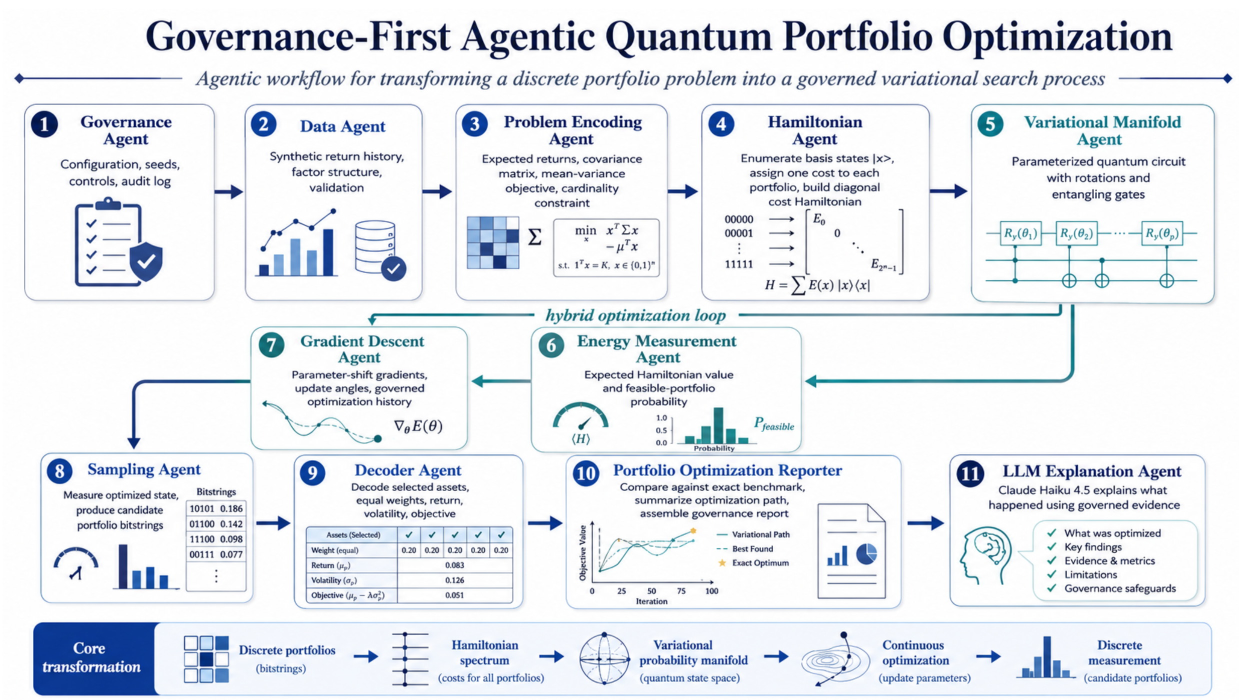

Figure 1: Governance-first agentic workflow for quantum portfolio optimization.

2 Why Agentic: Simplifying Complex Circuits and Governing Discovery

Quantum algorithms are compact by design. A circuit diagram can represent state preparation, encoding, entanglement, phase evolution, basis transformation, measurement, and classical post-processing in a small amount of space. This compactness is efficient for experts but often pedagogically difficult, especially for finance professionals who must understand not only whether a computation ran, but also whether the right economic question was encoded. A compact circuit can hide the causal sequence by which a financial problem becomes a quantum state and then becomes a portfolio again.

An agentic architecture expands the circuit into specialized computational responsibilities. An agent is not an anthropomorphic replacement for a gate and should not be understood as an unconstrained autonomous decision-maker. It is a bounded software role with a declared mandate, input contract, output contract, validation rules, and audit obligation. The Data Agent does not design the Hamiltonian. The Hamiltonian Agent does not change the return estimates. The Gradient Descent Agent does not read the exact optimal portfolio. The Reporter does not alter the measured bitstrings. This separation is analogous to model-development controls in quantitative finance: data engineering, model specification, estimation, optimization, validation, and reporting are related but distinct functions.

This separation reduces cognitive complexity. A learner can focus on one question at a time. What financial data are being used? How is the objective defined? Which bitstring represents each portfolio? How is the Hamiltonian constructed? What states can the circuit reach? How is energy measured? How are gradients estimated? How are final bitstrings decoded? Each agent answers one of these questions.

The architecture also highlights discovery as a process. The final portfolio does not appear in one step. It emerges through a chain of evidence. The data produce expected returns and covariance. These inputs produce an objective. The objective produces a Hamiltonian spectrum. The circuit produces a probability distribution. The Hamiltonian expectation produces a scalar energy. The gradient produces a navigation direction. Measurement produces candidate bitstrings. Decoding produces portfolio weights and financial metrics.

This stepwise structure is especially valuable in optimization because a poor result can arise for many reasons. The return data may be noisy. The penalty term may be too weak. The circuit may be under-expressive. The optimization landscape may be flat. The learning rate may be unstable. The sampled state may place insufficient probability on feasible portfolios. The decoder may misinterpret bit ordering. Agent boundaries make these failure modes easier to locate.

Governance is strengthened by explicit controls at each stage. The Data Agent verifies shape and finite values. The Problem Encoding Agent verifies covariance symmetry and positive semidefiniteness. The Hamiltonian Agent checks that every basis state receives a finite energy. The Variational Manifold Agent verifies state normalization. The Energy Agent checks that probabilities sum to one. The Gradient Agent records the complete optimization path. The Sampling Agent verifies shot totals. The Decoder validates cardinality and weight normalization. These checks convert the notebook from a demonstration into a controlled experiment whose intermediate claims can be challenged independently.

The use of an exact benchmark illustrates another governance advantage. For a six-asset educational problem, all 2^6 portfolios can be enumerated. The exact optimum is retained as a validation reference. However, it is not supplied to the variational optimizer. This prevents information leakage. The optimizer must navigate using only expectation values and gradients.

The LLM Explanation Agent is also carefully bounded. It receives the governed evidence after the computation is complete. It can explain the architecture, results, limitations, and failure modes. It cannot modify the statevector, optimization history, sampled counts, decoded portfolio, or benchmark comparison. The distinction between computation and narration is therefore explicit.

Agentic design also supports reuse and comparative research. A different Data Agent could import market data. A different Encoding Agent could implement transaction costs, tracking error, robust covariance estimates, or benchmark-relative risk. A different Hamiltonian Agent could construct a Pauli decomposition. A different Variational Agent could use QAOA, hardware-efficient ansatz layers, or problem-inspired mixers. A different Optimizer could use SPSA, COBYLA, natural gradients, or Bayesian optimization. Because the interfaces are explicit, components can be replaced without rewriting the complete workflow, and alternative designs can be compared under common governance evidence.

The most important conceptual contribution is that the agentic approach makes discovery governable. Rather than presenting a quantum result as a black-box answer, it records how each claim was produced. This is essential for financial institutions, where decisions must be traceable to data, assumptions, controls, and validation evidence.

3 Workflow I: Data, Financial Objective, and Quantum Encoding

The workflow begins with a synthetic return history designed to contain both common and idiosyncratic structure. The notebook uses a market factor, a sector factor, asset-specific expected returns, different market exposures, sector sensitivities, and idiosyncratic volatility. This creates a realistic covariance pattern without relying on external data.

The Data Agent records the random seed, number of observations, asset labels, sample means, and sample volatilities. Reproducibility is achieved through controlled pseudo-random generation. The purpose is not to claim realistic forecasting performance but to create a stable environment for examining the quantum methodology.

The Problem Encoding Agent annualizes expected returns and covariance:

$$\mu = 252\,\bar{r},$$

and

$$\Sigma = 252\,\mathrm{Cov}(r).$$

It then defines the portfolio cost. For equal-weight selected portfolios,

$$w_i = \frac{x_i}{K}.$$

The risk term is

$$w^\top \Sigma w,$$

and the expected return is

$$\mu^\top w.$$

The penalty

$$A\left(\sum_i x_i - K\right)^2$$

is central. Without it, the lowest-cost state might violate the cardinality requirement. The penalty transforms a constrained problem into an unconstrained energy objective. The value of A must be large enough to discourage infeasible states but not so dominant that it overwhelms the economic trade-off.

The Hamiltonian Agent enumerates all basis states. For $n = 6$, the dimension is

$$2^6 = 64.$$

Each index is converted into a six-bit string, and the corresponding portfolio cost is evaluated. The diagonal Hamiltonian is then represented by the vector

$$(C(000000), C(000001), \ldots, C(111111))\,.$$

The exact benchmark is the basis state with minimum energy. In a scalable implementation, complete enumeration would not be possible. Here it is retained for educational validation.

This stage demonstrates that the Hamiltonian is a scoring operator. It gives the portfolio set an ordered energy structure. Low-energy states represent attractive risk-return combinations that satisfy the cardinality requirement. High-energy states may represent poor financial trade-offs or constraint violations.

The Hamiltonian can also be expressed through Pauli Z operators. Since

$$x_i = \frac{1 - Z_i}{2},$$

a quadratic binary objective can be transformed into an Ising Hamiltonian containing terms such as

$$Z_i \quad \text{and} \quad Z_i Z_j.$$

The diagonal enumeration used in the notebook is pedagogically direct, while the Pauli form is more closely related to hardware execution.

The separation between financial encoding and Hamiltonian construction is important. The financial objective should be validated independently before it becomes a quantum operator. Otherwise, an error in the economic model can be mistaken for an error in the quantum circuit.

4 Workflow II: Hamiltonian, Variational Manifold, and Energy Landscape

The Variational Manifold Agent defines a layered circuit with trainable R_y rotations and a ring of controlled-NOT gates. Each rotation contributes a continuous parameter. The entangling gates create correlations among the qubits.

The initial state is

$$|0 \cdots 0\rangle.$$

After applying the circuit,

$$|\psi(\theta)\rangle = U(\theta)|0 \cdots 0\rangle.$$

The state contains one amplitude for each portfolio:

$$|\psi(\theta)\rangle = \sum_x \alpha_x(\theta)|x\rangle.$$

The circuit defines a map

$$\theta \longmapsto |\psi(\theta)\rangle.$$

The image of this map is the variational manifold. It is not the complete Hilbert space. It is the subset that can be reached with the chosen architecture and depth.

This restriction is both useful and dangerous. It is useful because a small number of parameters provides a compact coordinate system over an exponentially large basis. It is dangerous because the optimum may lie outside the reachable set.

The Energy Measurement Agent calculates

$$E(\theta) = \langle \psi(\theta) | H_C | \psi(\theta) \rangle.$$

For the diagonal Hamiltonian,

$$E(\theta) = \sum_x |\alpha_x(\theta)|^2 C(x).$$

The energy is therefore an aggregate signal over the full probability distribution.

The notebook also measures feasible probability:

$$P_{\text{feasible}}(\theta) = \sum_{x:\, \sum_i x_i = K} |\alpha_x(\theta)|^2.$$

This quantity is important because a state can have low expected energy while still assigning significant probability to invalid portfolios if the penalty is weak. For a financial audience, this is the analogue of monitoring both objective improvement and mandate compliance: a low reported loss is not acceptable if it is achieved by portfolios the mandate would not permit.

The entropy of the distribution is also recorded:

$$S(\theta) = -\sum_x P_\theta(x) \log P_\theta(x).$$

A high-entropy state spreads probability broadly. A low-entropy state concentrates probability on fewer candidates. Optimization often reduces energy by shifting probability toward a smaller family of attractive portfolios.

The parameter landscape

$$E(\theta)$$

is the continuous object navigated by the optimizer. It can contain valleys, ridges, plateaus, symmetries, and local minima. The shape is determined jointly by the Hamiltonian and circuit architecture.

The Hamiltonian does not create the manifold. The circuit creates the manifold. The

Hamiltonian assigns altitude to each point on it. This distinction is fundamental:

$$\text{circuit} \to \text{reachable states},$$

$$\text{Hamiltonian} \to \text{energy assigned to those states}.$$

The method therefore converts the original combinatorial problem into a continuous optimization over circuit coordinates. It does not eliminate combinatorial difficulty; it changes its form.

5 Workflow III: Gradient Descent, Sampling, Decoding, and Reporting

The Gradient Descent Agent estimates derivatives using the parameter-shift rule. For each parameter θ_j, it evaluates the energy at two shifted settings:

$$E\left(\theta + \frac{\pi}{2} e_j\right) \quad \text{and} \quad E\left(\theta - \frac{\pi}{2} e_j\right).$$

The gradient component is

$$\frac{\partial E}{\partial \theta_j} = \frac{1}{2}\left[E\left(\theta + \frac{\pi}{2} e_j\right) - E\left(\theta - \frac{\pi}{2} e_j\right)\right].$$

The parameters are updated through

$$\theta^{(t+1)} = \theta^{(t)} - \eta \nabla E(\theta^{(t)}).$$

The notebook uses gradient clipping and a governed learning-rate reduction when a proposed step increases energy. The complete history is retained, including unsuccessful proposals. This is important because optimization traces are evidence: they reveal sensitivity to initialization, learning-rate instability, plateaus, and possible overconcentration of probability mass.

The optimization history includes expected energy, feasible probability, entropy, gradient norm, and learning rate. This allows a reviewer to determine whether the algorithm is improving the economic objective, satisfying constraints, concentrating probability, and converging.

After optimization, the Sampling Agent draws repeated measurements from

$$P_{\theta^\star}(x) = |\alpha_x(\theta^\star)|^2.$$

Each shot returns one discrete bitstring. The frequency of a bitstring estimates its measurement probability.

The Sampling Agent does not automatically choose the most probable state. It filters for feasible portfolios and ranks them by observed frequency. This is important because the most probable raw state may violate the cardinality constraint.

The Decoder Agent converts the selected bitstring into an asset list and equal weights. It independently recomputes expected return, volatility, objective value, and constraint violation. It also verifies that the weights sum to one and that the objective matches the Hamiltonian energy assigned to that basis state.

The Reporter Agent compares the decoded portfolio with the exact benchmark. It calculates the optimality gap:

$$\mathrm{Gap} = C(x_{\mathrm{candidate}}) - C(x_{\mathrm{benchmark}}).$$

A zero gap indicates that the variational search sampled the exact optimum. A positive gap quantifies the difference.

The report also distinguishes optimization success from measurement success. The final state may assign substantial probability to the optimal portfolio without producing it in a small number of shots. Conversely, a good candidate may be sampled even when the state remains broad. This distinction is essential in probabilistic algorithms: the trained distribution, the finite-shot sample, and the final selected portfolio are related but not identical objects.

The governance report assembles configuration, financial inputs, Hamiltonian range, circuit parameters, optimization history, sampling evidence, decoded portfolio, benchmark, optimality gap, and audit records. This is the final governed artifact.

The LLM Explanation Agent then produces a natural-language account of what occurred. It is explicitly instructed to state that the notebook is an exact NumPy simulation and does not establish quantum advantage.

6 Applications to Quantum Computing Ideas in Financial Algorithms

The agentic methodology can be extended to many areas of quantum finance, provided that claims are benchmarked against strong classical baselines and that state-preparation costs are included in the analysis. The core pattern remains:

$$\text{data} \to \text{encoding} \to \text{quantum state} \to \text{measurement} \to \text{classical validation.}$$

In quantum amplitude estimation, a Data Agent can define a probability distribution, a Payoff Agent can encode a financial payoff, an Estimation Agent can execute phase-based amplitude estimation, and a Validation Agent can compare the result against classical Monte Carlo. Applications include option pricing, expected credit loss, counterparty exposure, and insurance risk.

In quantum risk analysis, agents can separately govern marginal distributions, dependence structure, scenario generation, loss encoding, tail probability estimation, and stress testing. This separation is particularly valuable because errors in dependence assumptions can dominate the final risk measure.

For derivative pricing, one agent can construct the stochastic process, another can discretize the state space, another can encode the payoff, another can execute the circuit, and another can estimate pricing error. The governance record should disclose discretization bias, state-preparation cost, and measurement uncertainty.

Credit portfolio analysis can use a similar decomposition. A Data Agent estimates default probabilities and correlations. An Encoding Agent maps defaults into qubits. A Loss Agent defines exposure and recovery. An Amplitude Estimation Agent estimates expected loss or tail loss. A Capital Agent translates the output into economic or regulatory capital.

Quantum machine learning can also adopt the architecture. Separate agents can govern feature selection, data encoding, circuit design, training, validation, fairness, and explanation. This is important because quantum machine-learning claims can be distorted by small sample sizes, weak classical baselines, or excessive preprocessing.

Hybrid quantum-classical trading systems could assign one agent to market-state encoding, another to signal Hamiltonian construction, another to circuit execution, another to portfolio translation, and another to transaction-cost and risk validation. The final trading decision should remain governed by classical constraints.

The approach is also relevant to scientific discovery in finance. An agent can propose a hypothesis, another can generate a quantum representation, another can test the model, another can compare classical baselines, and another can preserve the evidence. This transforms quantum experimentation into a reproducible research process.

The common principle is that quantum financial algorithms should not be treated as isolated computational kernels. They should be designed as governed systems with explicit responsibilities and evidence at every stage. For sophisticated users, the right question is not simply whether a quantum subroutine is elegant, but whether the complete pipeline improves decision quality after accounting for data quality, encoding overhead, sampling error, classical alternatives, operational risk, and model governance.

7 Conclusion

The governance-first agentic notebook provides a pedagogical architecture for understanding quantum portfolio optimization as a sequence of controlled transformations. The original financial problem is discrete and combinatorial. Each portfolio is represented by a bitstring, and the number of alternatives grows exponentially with the number of assets.

The Hamiltonian embeds the portfolio objective into an energy spectrum. Each basis state represents one candidate portfolio, and each eigenvalue represents its risk-return-penalty score. The variational circuit then creates a lower-dimensional family of probability distributions over the complete candidate set.

The expected Hamiltonian value

$$E(\theta) = \sum_x P_\theta(x) C(x)$$

becomes a continuous surrogate objective. Gradient descent navigates the circuit parameter space, while the circuit reshapes the probabilities of many portfolios simultaneously.

The final state is not itself the portfolio. Measurement converts the optimized distribution into discrete bitstrings. The decoder then translates those bitstrings into asset selections and validates the financial metrics.

The agentic architecture makes every stage explicit. Data, objective definition, Hamiltonian construction, circuit design, energy measurement, gradient descent, sampling, decoding, benchmarking, and explanation are treated as separate responsibilities. This supports transparency, reproducibility, and fault isolation.

The method also clarifies the limits of variational quantum optimization. A compact representation does not guarantee an efficient search. The optimal portfolio may lie outside the manifold. Gradients may vanish, local minima may dominate, and finite-shot estimates may be noisy. The state may remain broad, sampling may require many shots, and hardware noise may distort the result. Practical quantum advantage is therefore not assumed; it would require evidence against carefully selected classical baselines under comparable information and cost assumptions.

Nevertheless, the methodology provides a powerful educational and architectural framework. It shows how a massive discrete problem can be represented through a structured quantum state, how a Hamiltonian supplies an objective landscape, and how continuous parameters can navigate a probability distribution over combinatorial alternatives.

For financial institutions, this governance-first perspective is essential. Future quantum algorithms will need to satisfy the same standards applied to other decision models: documented data, transparent assumptions, reproducible execution, independent validation, benchmark comparison, and explainable outputs.

The notebook therefore serves as both a teaching instrument and a prototype for governable quantum-financial systems. Its central lesson is that quantum optimization should be understood not as an opaque search for a magical answer, but as a structured process of encoding, transformation, measurement, inference, and validation.

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